`

AL FLASHER 24 V TEMPERATURE AND HUMIDITY TEST REPORT

Part Name	AL FLASHER 24 V	**Application**	COMMERCIAL APPLICATION
Customer Part No.	FGG00700	**Customer**	ASHOK LEYLAND
UCAL Part No.	RDEPL/AL/16	**Report No**	R&D/AL/24VF/VAL/26/007

INTRODUCTION:

UCAL is currently supplying 24V AL Flashers to Ashok Leyland for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

PURPOSE:

To evaluate the 24 V flashers assembly after **Temperature and Humidity Test**

TEST SPECIFICATION:

Place the Electronic Component in the constant temperature and humidity chamber. Set the chamber inside temperature to 23±5°C and humidity to (60±15) %.Keep the Electronic Component under this condition for 2.5±0.5 hours. Carry out the test pattern shown in Fig 13 for 10 cycles and operate the Electronic Component for the specified time. Subsequent operations should be as follows: After keeping the test Electronic Component under the condition of temperature 25±5°C and humidity (65±20) % for 22±2 hours conduct the functional and visual inspection. After completion of the complete cycle conduct the functional and visual inspection without removing the water drops from the surface.(29Hrs/cycle)

ACCEPTANCE CRITERIA:

All function of the flasher performs as designed before and after the test**.**

RESULT:

SAMPLE NO	PERFORMANCE	APPEARANCE	JUDGEMENT
S:06	O	O	O

O: OK **X: NG** **∆: To be confirmed**

CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

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Prepared By	**Approved By**

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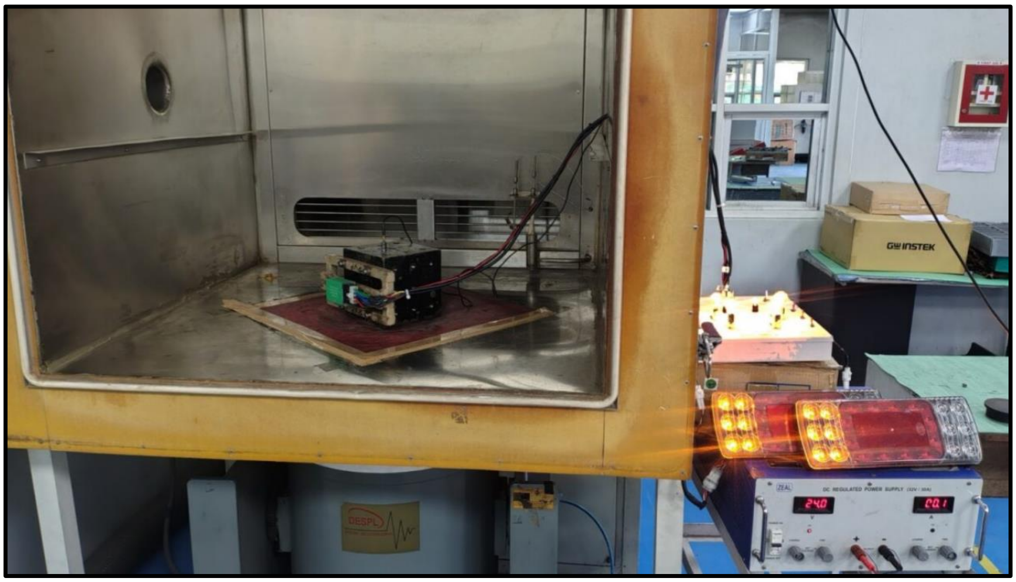
DATE:29.04.2026

PERFROMACE TEST SPECIFICATIONS:

SNO	PARAMETERS	SPECIFICATIONS
1	Operating voltage	24 ~ 32 V
2	Test voltage	24 V
3	Flash rate(Turn signal/Hazard warning	60~120 Flashes/Minute
4	One lamp failure flashing rate	>140 Flashes/Min(or)1.8 times of flash rate whichever is greater
5	Duty Cycle	40% ~60%

TEST SETUP:



TURN SIGNALS & HAZARD SIGNALS:

LEFT SIGNAL	RIGHT SIGNAL	HAZARD SIGNAL
		

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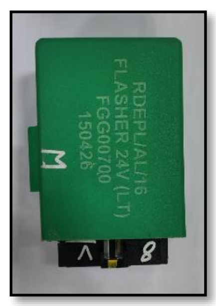
TEST OBSERVATION:

Description	Load	Observations(at 12 V)	Remark
	LED& CLUSTER	Sample No:01	
Turn Signal RH	2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal	85	OK
Turn Signal LH	2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal	85	OK
Hazard Mode	2x (2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal)	85	OK
Fault condition Any one 21W Lamp failure	2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal	171	OK
Duty cycles for RH&LH	2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal	49%	OK
Duty cycles for Hazard Mode	2x (2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal)	49%	OK

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

TESTED SAMPLES PHOTOS:

BEFORE TEST SAMPLE – S:07	AFTER TEST SAMPLE – S:07
	

OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed before & after the test.